

Deep Learning with Differential Privacy

The Experiment

- Write the code to implement the Differential Privacy method applied in Deep Learning
 - You can use the method described in this paper:
[M. Abadi, et al. "Deep learning with differential privacy." *Proceedings of the 2016 ACM SIGSAC conference on computer and communications security*. 2016.](#)
- You can use the dataset mentioned in this paper.
- Then, run the “Deep Learning with Differential Privacy” model on the above dataset, and observe how the privacy protection impacts on the machine learning system.
- Finally, provide the results
 - How do you perturb the data?
 - Effectiveness measure: Accuracy
 - **Privacy level: the value of ϵ**
- Programming Language is not limited.

Reference (Coding)

- [GitHub - lingyunhao/Deep-Learning-with-Differential-Privacy: An implementation of Deep Learning with Differential Privacy](#)
 - Mainly use Google's *TensorFlow Privacy*
 - [Introducing TensorFlow Privacy: Learning with Differential Privacy for Training Data — The TensorFlow Blog](#)
 - [GitHub - tensorflow/privacy: Library for training machine learning models with privacy for training data](#)

Reference (Blog)

- [Deep Learning with Differential Privacy \(DP-SGD Explained\) \(mukulrathi.com\)](https://mukulrathi.com/2018/05/28/deep-learning-with-differential-privacy-dp-sgd-explained/)
- [Deep Learning with Differential privacy \(Mandrian\)](https://www.mandrian.io/blog/2018/05/28/deep-learning-with-differential-privacy/)

Upload the result to Moodle

- Upload your code and the *detailed* reports to Moodle.
 - Code: You must add some comments in your code for easy understanding.
 - Report Files: Document (Word) and Presentation (PPT).
 - Including a table in your report outlining the responsibilities of each team member for this final project.
 - **Video: You should record a video of the presentation, and then provide the video file, or an online address (e.g., Youtube).**
 - Zipped to a file. The file name should be “Team Name_Midterm.zip”